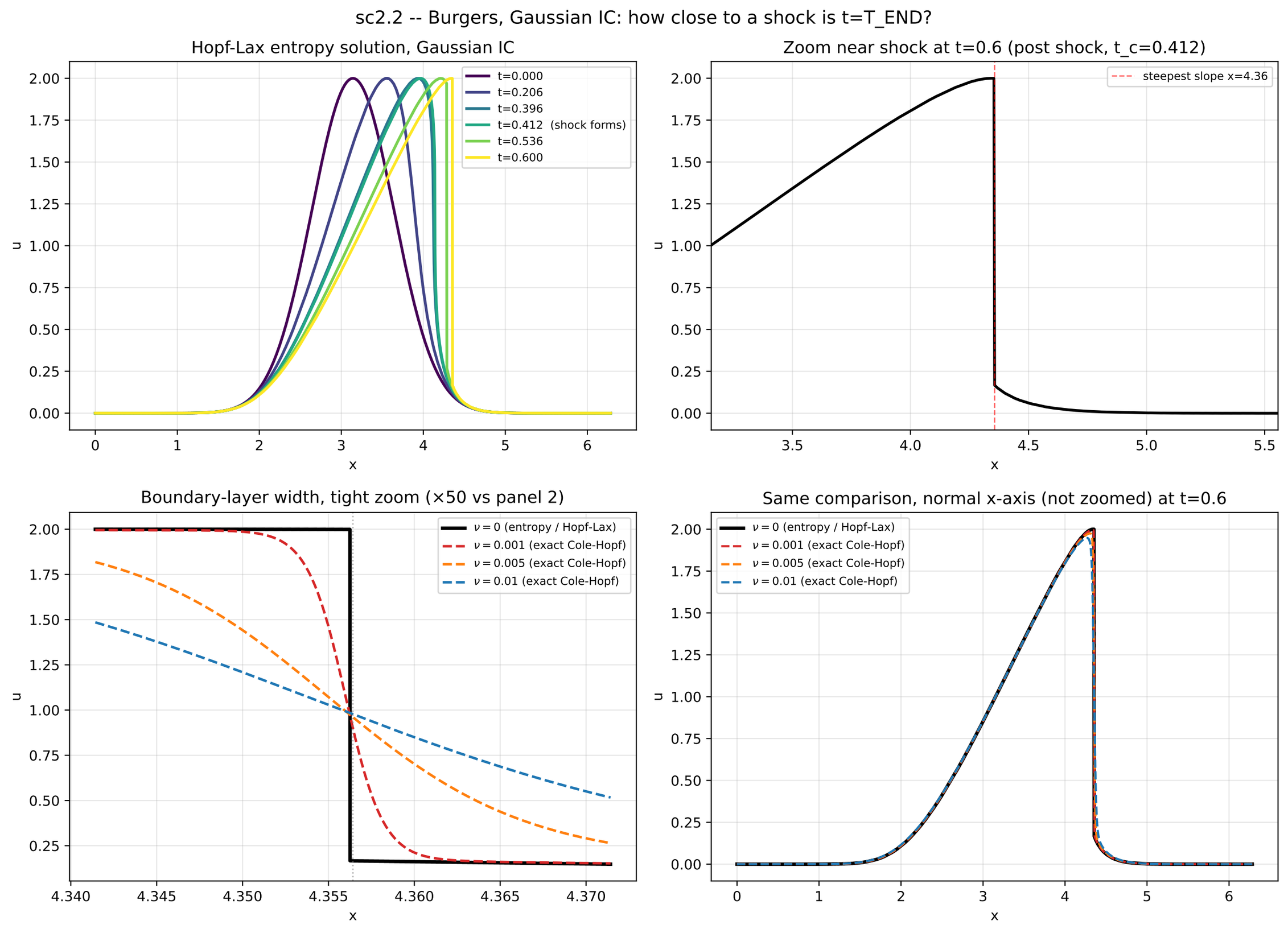
sc2.2 -- Burgers, Gaussian IC: how close to a shock is t=T_END?
Hopf-Lax entropy solution, Gaussian IC
t=0.000
t=0.206
t=0.396
t=0.412 (shock forms)
t=0.536
t=0.600
u
x
Zoom near shock at t=0.6 (post shock, t_c=0.412)
steepest slope x=4.36
u
x
Boundary-layer width, tight zoom (×50 vs panel 2)
ν = 0 (entropy / Hopf-Lax)
ν = 0.001 (exact Cole-Hopf)
ν = 0.005 (exact Cole-Hopf)
ν = 0.01 (exact Cole-Hopf)
u
x
Same comparison, normal x-axis (not zoomed) at t=0.6
ν = 0 (entropy / Hopf-Lax)
ν = 0.001 (exact Cole-Hopf)
ν = 0.005 (exact Cole-Hopf)
ν = 0.01 (exact Cole-Hopf)
u
x